Incremental Graph Layout by Stochastic Search: a Routed Objective, a Calibrated Metric, and a Matched-Budget Control

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Abstract

Placing the nodes of a graph in the plane with few edge crossings is NP-complete, so every layout tool runs heuristics, and the heuristics are rarely explicit about three things: what objective they optimize, whether that objective measures the drawing the reader actually sees, and whether the search spends its budget any better than uniform random sampling would. We treat all three. The problem is formulated incrementally: a graph grows by k nodes and the existing drawing is frozen, which keeps the reader's mental map, bounds the search space at $2k$ variables, and contains the classic full redraw as the limit of an anchor term that prices the displacement of frozen nodes. The objective reads the drawing's real medium, orthogonally routed connectors rather than the straight center-to-center segments most objectives score, and the metric is calibrated: the router's shortfall against a reference layout of certified on-screen quality is measured and printed beside every number that depends on it. Four families of stochastic search (hill climbing with restarts, simulated annealing, a genetic algorithm, and uniform random sampling as the matched-budget control) are compared by exact sign tests on paired seed panels, deterministically reproducible to the byte across platforms. The motivating application is the entity–relationship diagram, which mainstream tools still place badly enough that maintainers redraw by hand; a proof of concept places the tables added by eight real consecutive schema migrations of a production database, each with a human-accepted answer, under a pre-registered analysis. The searches beat the deployed-style centroid heuristic on every migration (exact Wilcoxon, Holm-corrected $p = 0.023$); hill climbing, not the genetic algorithm the pre-registration predicted, is the budget-efficient method (it wins every non-tied migration against the GA, $p = 0.016$, and separates from the control by 2000 evaluations on most pairs); and on the half of real migrations that add only one or two tables, no search separates from uniform sampling at all, so the first deployable rule is to count the new nodes before reaching for a search. One general finding: under any objective of this family the searches outscore the human reference, which measures the objective's blindness to what maintainers optimize rather than any superiority of the tool, and we argue this gap is why automatic graph layout has stayed bad in practice.

1 Introduction

[PENDING: introduction]

2 The problem

[PENDING: formulation]

*Independent researcher. Code, data and derived tables: `https://github.com/Anode1/cjitter`.

3 The objective is the medium

[PENDING: objective]

4 Metrics and statistics

The objective over a placement v of the k new nodes is

$$S(v) \;=\; 100\,P(v) \;+\; 100\,C(v) \;+\; L(v),$$

with P the total length of routed connector lying inside a node it does not join, C the count of proper crossings between routed connectors of disjoint node pairs, and L the total routed length; the weights are tiers two orders of magnitude apart, so length only breaks ties and no weight is tuned. The same S is defined for the straight edge model by replacing the router with the direct segment. Beside S:

- the *feasibility pair* (C, P), ordered lexicographically: the part of the score a reader can verify by looking at the drawing;
- the *calibration* $\chi = (C_{\mathrm{ref}}, P_{\mathrm{ref}})$, the metric applied to a reference layout whose on-screen quality is certified as zero and zero. The gap between χ and $(0, 0)$ is the metric's own error, printed beside every number that depends on it;
- the *transfer gap* $\Delta = S_{\mathrm{routed}}(\hat{v}_{\mathrm{straight}}) - S_{\mathrm{routed}}(\hat{v}_{\mathrm{routed}})$: the routed cost of having optimized the straight model, what scoring the wrong medium costs;
- the *separation budget* B^*: the smallest evaluation budget at which a method's exact one-sided sign test against the matched-budget control reaches $p \le 0.05$ on the shared seed panel. A method that never separates gets $B^* = \infty$, a verdict the layout literature does not risk because it does not run the control.

Within an instance, methods share one seed panel and are judged by the exact one-sided sign test on paired per-seed differences. Across instances, the pre-registered comparisons use the exact Wilcoxon signed-rank test with Holm correction over the declared family, Hodges– Lehmann estimates with distribution-free intervals for effect sizes, and a minimum detectable effect wherever a null is reported, so a null bounds rather than asserts. Realized: $n = 8$ of 16 consecutive pairs (seven added no tables, one was the pilot), five seeds per method per pair, RESULTS.md beside this file carrying the full battery.

5 The searches and the control

[PENDING: library]

6 The benchmark

[PENDING: benchmark]

7 Pre-registered experiment

[PENDING: results]

8 What the human comparison measures

[PENDING: discussion]

9 Related work

[PENDING: related]

10 Conclusion

[PENDING: conclusion]